Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/22/2008
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Neurochemical and Behavioral Effects of Exposure to Heavy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rabin@umbc.edu	**Fax:**	FY (410) 455-1055
PI Organization Type:	UNIVERSITY	**Phone:**	(410) 952-1761
Organization Name:	University of Maryland, Baltimore County		
PI Address 1:	Department of Psychology		
PI Address 2:	1000 Hilltop Cir		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Sept 2007--Received NCE to 12/31/2008 per S. Krenek/JSC; original end date was 12/31/2007		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Joseph, James (USDA, HNRCA at Tufts University) Shukitt-Hale, Barbara (USDA, HNRCA, Tufts University)		
Grant/Contract No.:	NNJ06HD93G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars) will involve long-term travel beyond the magnetic field of the earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles and particles of high energy and charge (HZE particles). Previous research has shown that exposure to HZE particles can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements. The objectives of the proposed research program are: (1) to further characterize HZE particle-induced behavioral and neurochemical deficits; (2) to determine the neural mechanisms underlying the changes in performance; and (3) to determine how individual characteristics (age and gender) may influence the responsiveness to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Effects of Exposure to Different Particles: Based on our previous research with a range of energies of 56Fe particles, we had assumed that the acute effects of exposure to a variety of HZE particles would show the LET-dependence such that decreasing particle LET would produce deceases in the relative biological effectiveness. The results of the research reported here does not completely support that assumption. Despite the fact that the LET of 28Si (600 and 1000 MeV/n; LET ˜51 and 44 keV/µm, respectively), 48Ti particles (LET ˜106 keV/µm) and 12C (LET ˜13 keV/µm) is significantly less than that of 600 and 1000 MeV/n 56Fe particles (LET ˜189 keV/µm and 149 keV/µm, respectively), the thresholds for the disruption of neural and behavioral functioning following exposure to these particles was significantly less than that following exposure to 56Fe particles. Additional research completed this year is consistent with the observation that lower LET particles produce a greater disruption of neurocognitive performance. Because these doses of 12C particles needed to disrupt neurocognitive performance were so low (5-10 cGy) are so low, it was important to get a measure of the reliability of these effects on neurocognitive performance. A second set of rats was irradiated with 12C ions and then tested with the same doses. These experiments showed that equivalent doses were needed to disrupt neurocognitive performance. It may also be noted that there is some variability in the absolute values of the thresholds for the original experiment and the replication. These differences may reflect differences in handling, in temperature, in the environment, as well as differences in the temperament of the individual subjects. Nonetheless, because the relative thresholds were similar the results of the replications confirm that the results obtained during the first series of experiments were reliable and that exposing rats to doses of HZE particles in the range to which astronauts will be exposed on long-duration missions can affect neurobehavioral performance. Also, this research also indicated that irradiation with different particles and energies did not produce consistent decrements in all measured endpoints. As such, the specific neural and behavioral endpoint may also be a significant factor in determining the effects of exposure to HZE particles. The data obtained thus far would suggest that it may not be possible to determine the acute effects of exposure to the range of HZE particles to which astronauts may be exposed on exploratory class missions by simply studying the effects of exposure to a single particle or measuring a single endpoint. Additional research will be needed to determine the nature of the relationship between exposure to different heavy particles and exposure-induced changes in neurobehavioral endpoints. Effects of age: Previous research had shown that exposure to a dose of 150 cGy of 56Fe particles did not disrupt the ability of the rat to respond appropriately on an ascending fixed-ratio schedule when tested several months following irradiation. However, when tested 12 months following exposure, these rats showed a disruption of operant responding. Similar effects were obtained following exposure to other HZE particles, including 28Si, 48Ti and 12C. As part of the current sequence of experiments, the interaction between exposure to HZE particles and aging was evaluated using object recognition memory. As observed with the operant responding task, exposing rats to doses of HZE particles that do not affect performance on the object recognition task three months following irradiation, do affect the performance of rats when they are tested again eleven months following irradiation. Although additional research using even lower doses of these particles is needed to confirm these results, the present results are consistent with the hypothesis that there is a continuing performance decrement following exposure to HZE particles as a function of the time following irradiation. Radiation-Induced Changes in Neural Structure: The final series of experiments evaluated the effects of exposure to 150 cGy of 56Fe (1000 MeV/n) particles. This dose was selected because it produces significant decreases in spatial learning and memory, a task which is dependent upon the integrity of the hippocampus. Exposing rats to 56Fe particles produced a significant decrease in dendritic branching in Layers II and III of the cortex, but did not affect the number of dentritic spines in the cortex. However, irradiation did cause a significant decrease in the number of M-type (mushroom-type) spines on the dendrites of granule cells in the dentate gyrus of the hippocampus. The results of this experiment indicate that there are changes in the expression of dendrites in the hippocampus as a function of exposure to HZE particles and these changes are related to changes in performance.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Rabin BM, Joseph JA, Shukitt-Hale, B, Carey AN. "Dietary modulation of the effects of exposure to 56Fe particles." Advances in Space Research, 2007;40(4):576-80. http://dx.doi.org/10.1016/j.asr.2007.05.029 , Oct-2007
Articles in Peer-reviewed Journals	Carrihill-Knoll KL, Rabin,BM, Shukitt-Hale B, Joseph JA, Carey A. "Amphetamine-induced taste aversion learning in young and old F-344 rats following exposure to 56Fe particles." AGE 2007 Sep;29(2-3):69-76. http://dx.doi.org/10.1007/s11357-007-9032-1 , Sep-2007
Articles in Peer-reviewed Journals	Shukitt-Hale B, Casadesus G, Carey AN, Rabin BM, Joseph JA. "Exposure to 56Fe irradiation accelerates normal brain aging and produces deficits in spatial learning and memory." Adv Space Res. 2007;39(6):1087-92. http://dx.doi.org/10.1016/j.asr.2006.11.005 , May-2007
Books/Book Chapters	Rabin BM, Joseph JA, Shukitt-Hale B. "Effects of Berry Fruits on Neurocognitive Deficits Produced by Exposure to Space Radiation." in "Herbals for Radioprotection and Radiosensitization." Ed. R. Arora. Wallingford, United Kingdom : CABI Publishing, In Press, 2008. Expected publication August 2008., Jan-2008